ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2021
2 hours 30 minutes
PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

Additional materials: Mathematical tables or a non-programmable calculator may be used.
INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. (a) Expand and simplify $(3x - 1)(x + 1)$. [3]<<
>>(b) Factorise completely $20x^2 + 4x$. [2]<<
>>(c) Solve $3(x - 1) = 15$. [3]<<
>>(d) Simplify $(x^2 - 4)/(x - 2)$. [2] **[10]**

2. AB is parallel to CD. Transversal PQ meets AB at X and CD at Y.<<
>>(a) Angle AXY = 40°. Find the corresponding angle at Y. Give a reason. [2]<<
>>(b) Find the co-interior (allied) angle to 40°. Give a reason. [2]<<
>>(c) In triangle PQR, $\hat{P} = 40°$ and $\hat{Q} = 55°$. Calculate $\hat{R}$. [2]<<
>>(d) State whether triangle PQR is acute, right-angled or obtuse. Give a reason. [2] **[8]**

3. A rectangle measures 18 cm by 8 cm. A border of width 2 cm is drawn around the outside.<<
>>(a) Find the area of the rectangle. [2]<<
>>(b) Find the area of the outer rectangle. [3]<<
>>(c) Hence find the area of the border. [3] **[8]**

4. The line L has equation $y = 3x + 4$.<<
>>(a) Write down the gradient of L and the coordinates of the y-intercept. [2]<<
>>(b) Find the coordinates of the point where L meets the x-axis. [2]<<
>>(c) Find the gradient of the line joining A(1, 5) and B(3, 7). [2]<<
>>(d) Write the equation of the line parallel to L that passes through (0, 1). [2] **[8]**

5. The table shows values of x with frequencies f:<<
>>x=2, f=2, x=3, f=5, x=5, f=7, x=5, f=8, x=7, f=2.<<
>>(a) State the modal value of x. [1]<<
>>(b) Calculate Σf and Σfx. [3]<<
>>(c) Calculate the mean of x. [3]<<
>>(d) Find the range of x. [2] **[9]**

6. A = (4 0 ; 2 1) and column vector u = (2 ; 5).<<
>>(a) Calculate Au. [3]<<
>>(b) Find det A. [2]<<
>>(c) Describe fully the transformation represented by (0 −1 ; 1 0). [2]<<
>>(d) Find the image of the point (2, 5) under a reflection in the y-axis. [2] **[9]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

7. (a) Factorise $x^2 - 6x + 5$. [3]<<
>>(b) Hence solve $x^2 - 6x + 5 = 0$. [2]<<
>>(c) Write down the roots of $(x - 1)(x - 5) = 0$. [1]<<
>>(d) Sketch $y = (x - 1)(x - 5)$, showing the intercepts with both axes. [3]<<
>>(e) State the coordinates of the turning point. [3] **[12]**

8. In triangle ABC, $\hat{C} = 90°$, $\hat{A} = 30°$ and AC = 20 cm (hypotenuse).<<
>>(a) Write down sin 30° and cos 60°. [2]<<
>>(b) Calculate BC (opposite 30°). [3]<<
>>(c) Calculate AB (adjacent to 30°). [3]<<
>>(d) Show that $\tan 30° = 1/\sqrt{3}$ and hence find BC/AB. [4] **[12]**

9. O is the centre of a circle. A, B and C lie on the circumference. AB is a diameter. Angle BAC = 44°.<<
>>(a) Find angle ACB. Give a reason. [3]<<
>>(b) Find angle ABC. [2]<<
>>(c) Find angle BOC (angle at the centre standing on arc BC). Give a reason. [4]<<
>>(d) Point D lies on the remaining circumference. Find angle BDC. Give a reason. [3] **[12]**

10. The position vectors of A and B are a = (5 ; 3) and b = (0 ; 6). M is the midpoint of AB.<<
>>(a) Find vector AB. [3]<<
>>(b) Find |AB|. [3]<<
>>(c) Find the position vector of M. [3]<<
>>(d) Show that AM = ½ AB. [3] **[12]**

11. 40 candidates sat a Mathematics paper marked out of 50. A cumulative frequency curve (ogive) is drawn.<<
>>(a) State the median position on the cumulative-frequency axis. [2]<<
>>(b) Describe how to read the median mark from the ogive. [3]<<
>>(c) The lower quartile is at position $n/4$ and the upper quartile at $3n/4$. Write these positions as numbers. [3]<<
>>(d) Define the interquartile range and state one advantage of the IQR over the range. [4] **[12]**

12. Triangle PQR has P(2, 1), Q(4, 1) and R(2, 4).<<
>>(a) P' is the image of P under reflection in the y-axis. Write down the coordinates of P'. [2]<<
>>(b) Q is translated by the vector (2 ; -1). Find the image Q''. [3]<<
>>(c) The matrix (0 −1 ; 1 0) represents a transformation T. Describe T fully. [4]<<
>>(d) Find T(Q), the image of Q under T. [3] **[12]**

13. (a) y varies directly as x. When x = 6, y = 18.<<
>>   (i) Find the equation connecting y and x. [3]<<
>>   (ii) Find y when x = 8. [2]<<
>>(b) z varies inversely as x. When x = 3, z = 8. Let this value of z be z_0.<<
>>   (i) Find z when x = 6. [4]<<
>>   (ii) Describe what happens to z when x is doubled. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).